

RideShield: Shift-Based Micro-Insurance for Gig Workers

Domain : FinTech (Insurance & Gig Worker Protection)

Category: Software Edition

Team ID: SIH26_11

Team Name: Team Ithaca

Problem Statement

THE PROBLEM

- **Variable exposure:** Gig income and road exposure vary shift-by-shift and day-to-day.
- **Coverage mismatch:** Many conventional insurance products use fixed coverage periods rather than shift-based exposure.

AFFECTED USERS & STAKEHOLDERS

Primary users

- Delivery riders
- Ride-hailing drivers

Key stakeholders

- Gig platforms
- IRDAI-licensed insurers

EVIDENCE, SCOPED BY SOURCE

12M

gig workers, India, FY25, up 55% from 7.7M FY21

Economic Survey 2025-26 [1]

32.36%

food delivery workers reporting an on-job accident, n=173

Chennai food-delivery study [2]

No platform-provided paid leave or pension

surveyed platform workers, n=924, 28 cities

NCAER 2023 [3]

ROOT CAUSE : Protection models are not fully designed around the **variable, shift-based nature of gig work.**

[1] Economic Survey 2025–26, Government of India • [2] Chennai food-delivery study, IOSR (n=173) • [3] NCAER Platform Workers Report, 2023 (n=924)

Literature Survey & Existing Solutions

EXISTING SOLUTIONS

PLATFORM INSURANCE [11]

Example: Zomato / Swiggy
Provides: Accident & health cover
Limitation: Not dynamically priced per shift

UBI / TELEMATICS [10][5]

Example: IRDAI-approved motor add-ons
Provides: Usage + driving behaviour pricing
Limitation: Mainly motor-focused, not gig-worker accident cover.

WHAT RESEARCH SHOWS



UNDER-PROTECTION PERSISTS[4]
Gig workers remain under-protected.
Insurance and social-protection access gaps persist



RISK CAN BE PERSONALIZED [5][6]

Usage + behaviour can improve risk assessment and support risk-based pricing



CONTEXT MATTERS[7]

Traffic, weather and road conditions influence driving risk



AUTOMATED PAYOUTS ARE FEASIBLE[8]

Objective triggers can support automated insurance payouts



RELIABILITY IS THE CHALLENGE[6][9]

Data quality, privacy, false positives and validation remain important barriers

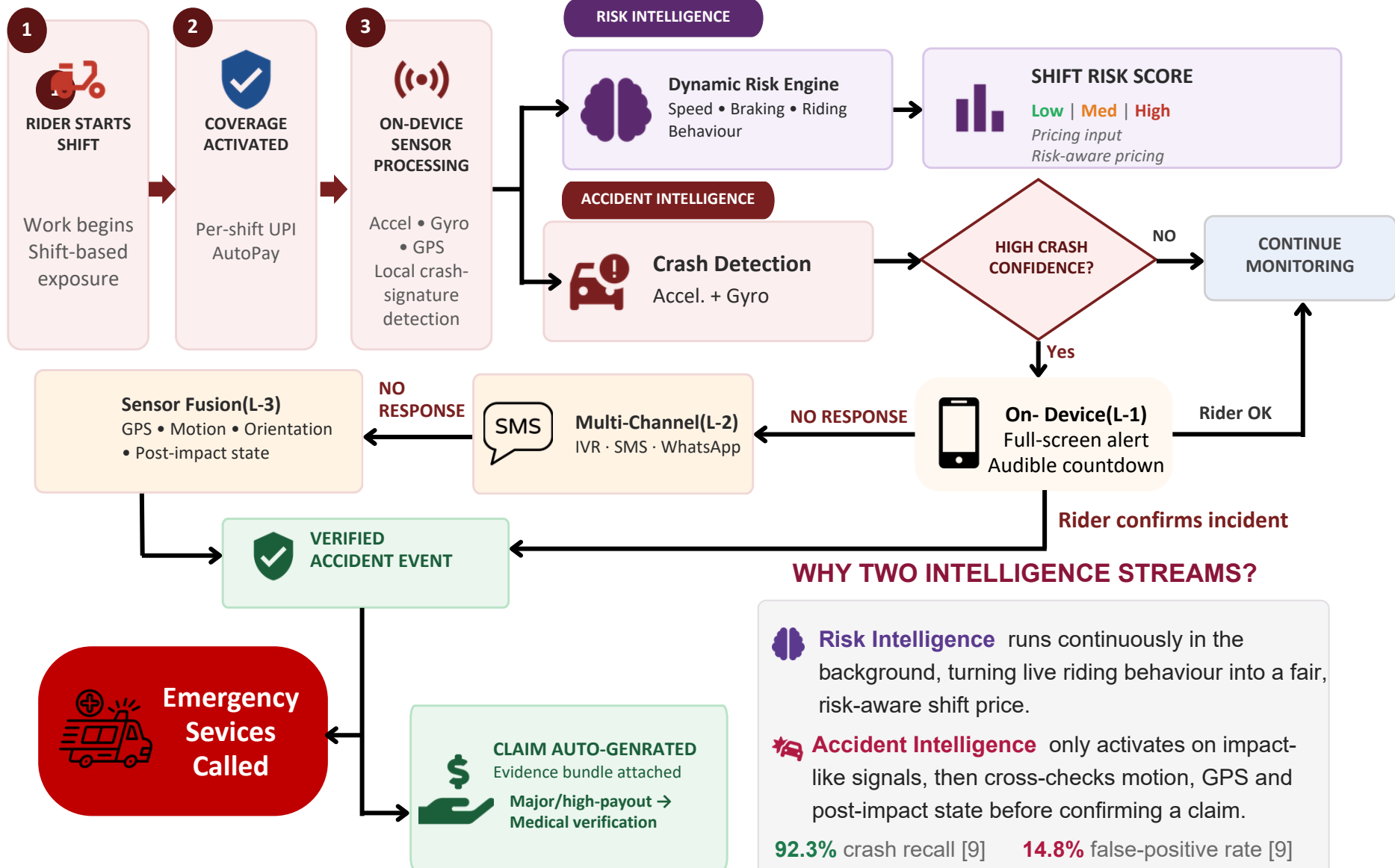
IDENTIFIED RESEARCH GAP



Existing solutions and research largely address these capabilities separately. We found limited evidence of their integration into an affordable, per-shift accident microinsurance model for gig workers.

Proposed Idea

HOW RIDESHIELD WORKS



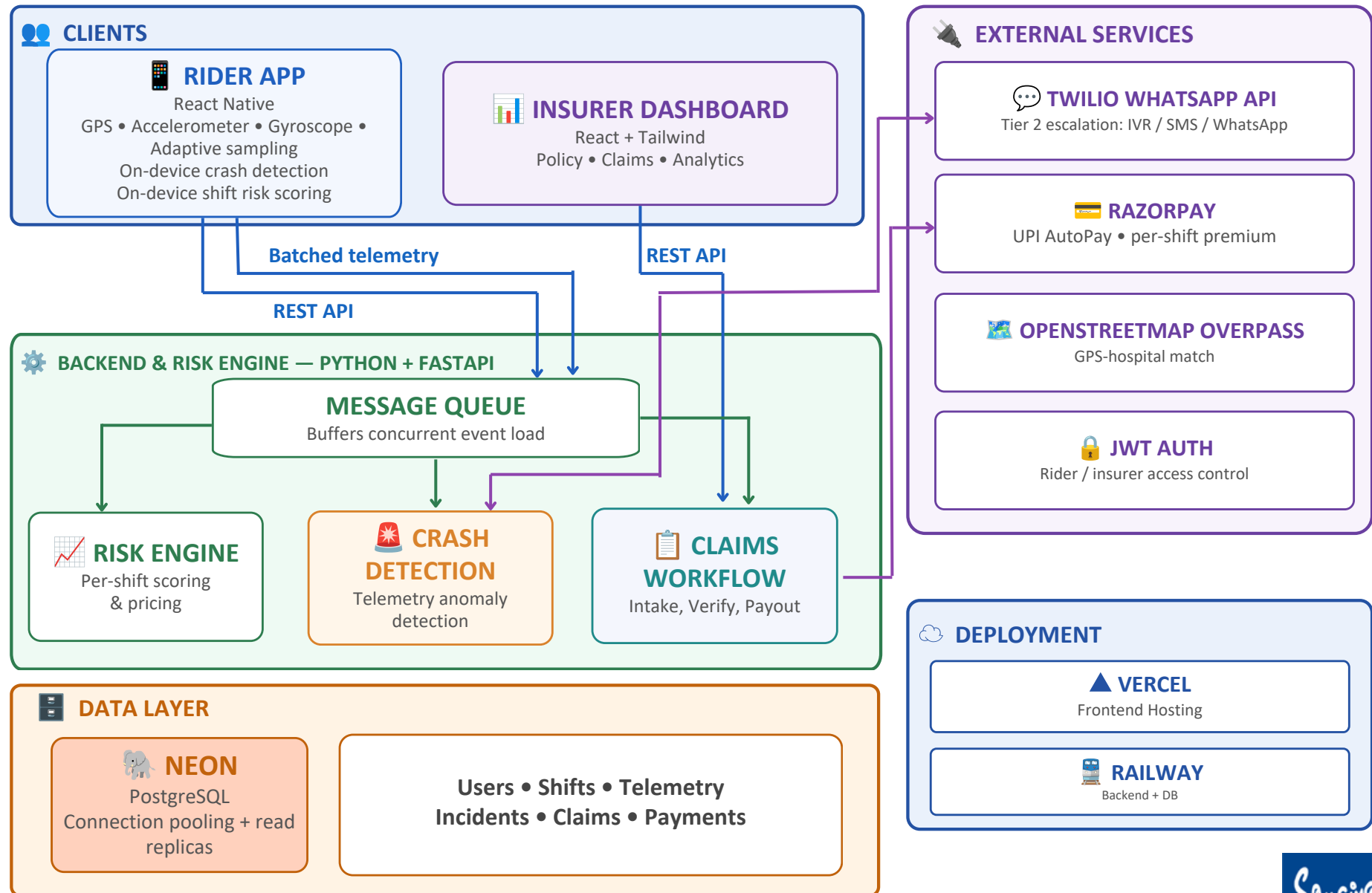
WHY TWO INTELLIGENCE STREAMS?

Risk Intelligence runs continuously in the background, turning live riding behaviour into a fair, risk-aware shift price.

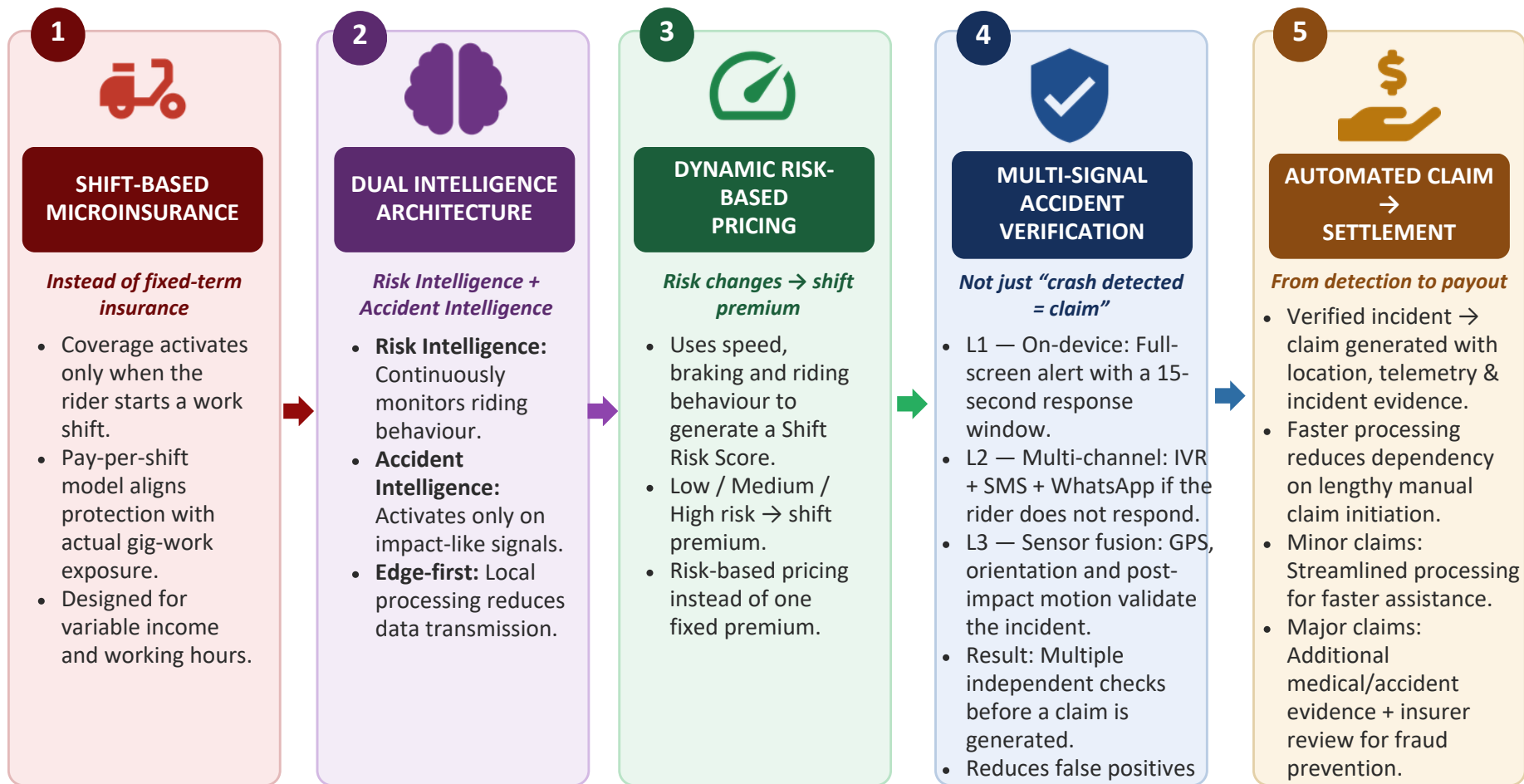
Accident Intelligence only activates on impact-like signals, then cross-checks motion, GPS and post-impact state before confirming a claim.

92.3% crash recall [9] **14.8%** false-positive rate [9]

Tech stack / Technology

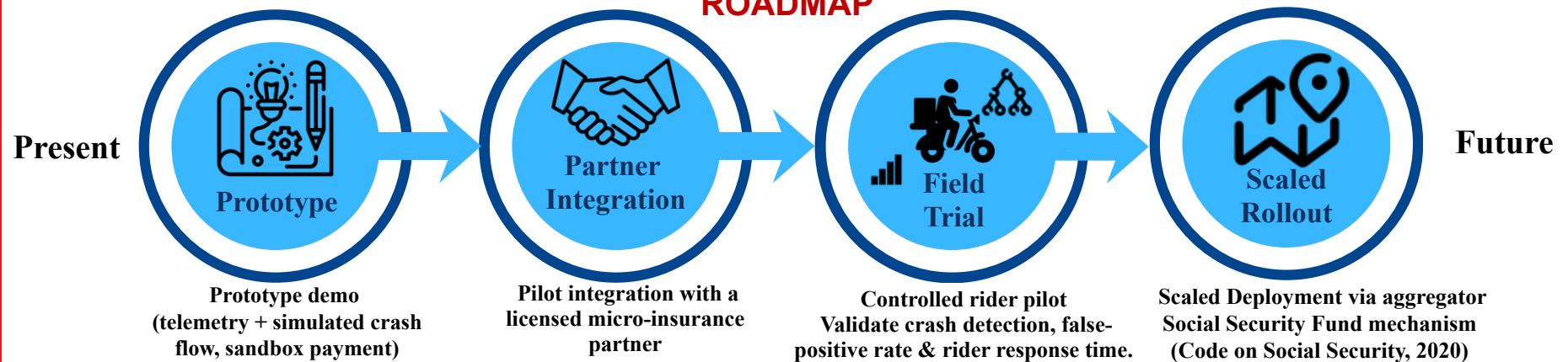


Innovation / Differentiation



RideShield does not introduce just another insurance product — it integrates shift-based coverage, dynamic risk assessment, accident verification and evidence-driven claim processing into one gig-worker protection ecosystem.

ROADMAP



FEASIBILITY

Built & Demonstrable

telemetry engine (GPS + accelerometer + gyroscope, live on-device), rule-based crash-signature filtering (multi-signal: accel spike + gyro + stationary window), UPI AutoPay flow (sandboxed), confirmation-before-claim step+on-device crash-signature + risk scoring (edge inference, not continuous streaming)

Future Scope

- Insurer integration
- Payout integration
- Large-scale field validation
- Load testing

EXPECTED REACH: 12M gig workers, India, FY25 [1] — RideShield's addressable population

RISK & MITIGATION

False-positive crash detection	Multi-signal verification + rider confirmation before claim generation; severity-based medical verification for major claims
No existing actuarial/pricing model for this population	RideShield positions itself as the risk-scoring + real-time pricing/collection layer, plugging into an existing IRDAI-licensed micro-insurance partner rather than inventing underwriting from scratch
Concurrency at scale (millions of concurrent riders/shifts)	Message queue (Kafka/SQS) decouples telemetry ingestion from processing; on-device edge inference reduces continuous server load per rider; DB connection pooling + read replicas handle concurrent read/write

References

- [1] Economic Survey 2025–26, Government of India
- [2] Chennai food-delivery study, IOSR (n=173)
- [3] NCAER Platform Workers Report, 2023 (n=924)
- [4] Sharma & Nuwal (2026). Legal Classification of Gig Workers in Rajasthan. J. Advances & Scholarly Research. IEEE / DOI
- [5] Cunha & Bravo (2022). Automobile UBI: Improving Risk Management via Telematics. CISTI, IEEE. IEEE / DOI
- [6] Qi et al. (2025). Telematics in Insurance: Challenges & Limitations. IEEE Access 13. IEEE / DOI
- [7] Masello et al. (2023). Contextual Data to Predict Risky Driving Events. Accident Analysis & Prevention 184. ScienceDirect / DOI
- [8] Rabehaja et al. (2024). Blockchain-Based Parametric Insurance. IEEE Trans. Services Computing 17(3). IEEE / DOI
- [9] Allewar (2026). Crash & Fall Detection for Delivery Riders via Smartphone Sensors. IJRASET. IJRASET / DOI
- [10] IRDAI (2022). Introduction of New Add-ons in Motor Insurance. Circular IRDAI/NL/CIR/Motor/136/07/2022. IRDAI Circular
- [11] Zomato. Delivery Partner Insurance Coverage. Zomato Delivery Partner
- [12] Bajaj Allianz (2025). ClimateSafe: Parametric Climate-Risk Insurance [Press release]. Bajaj Allianz Press Release

Team composition

Member no	Skills
Member 1	Mobile App Development • Flutter/Dart • GPS & Sensor Integration
Member 2	Crash Detection & Edge ML • Python • Sensor Fusion • On-Device ML
Member 3	UI/UX Design • Figma • Mobile/Web UX • Data Visualization
Member 4	Backend & API Development • Node.js/Express • REST APIs • Authentication
Member 5	Data Engineering • PostgreSQL • Redis • Data Processing
Member 6	Cloud & Security • Docker • AWS • CI/CD • API Security